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BAGONG PILIPINAS

CC 101 – Introduction to Computing

Laboratory Worksheet No. 3 | Week 6: Computer Networks & the Internet (CO4)

"Know Your Network Settings": Exploring Windows 11 Network Configuration Tools

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Computer Unit Used (Desktop/Laptop, brand): Laptop	**Date: October 16, 2026**

I. Objectives

At the end of this activity, you should be able to:

- Locate network adapter information using Windows 11 Settings, the Control Panel, Command Prompt (CMD), and PowerShell.
- Use **ipconfig**, **getmac**, **hostname**, **ping**, **netstat**, **route print**, and **systeminfo** in CMD, and **Get-NetAdapter**, **Get-NetIPAddress**, and **Test-Connection** in PowerShell.
- Interpret a sample **ipconfig /all** output from a networked computer: identify its IP address, subnet mask, default gateway, and DNS servers, and classify the address as public or private.
- Relate each tool and finding to the layer of the OSI model where it works.
- Design a simple network and IP address plan for the computer laboratory.

II. Materials Needed

- A computer running Windows 11 (a network or Internet connection is **not** required for this activity)
- This worksheet (printed copy, or the digital file to fill out and submit)
- Week 6 lecture notes, for reference on IP addresses, protocols, the OSI model, and network devices

III. Background

A computer does not need to be connected to a network to have network settings. Every PC has one or more **network adapters** (NICs), each with a permanent **MAC address**, and Windows always runs the TCP/IP software, including the **loopback** address 127.0.0.1 (IPv4) and ::1 (IPv6) that the computer uses to talk to itself. Windows 11 lets you inspect these settings in four places: the **Settings** app, the classic **Control Panel**, **Command Prompt (CMD)**, and **PowerShell**. In this activity, you will explore all four on your own unit, observe how Windows behaves when no network is available, and then analyze a sample configuration taken from a computer that *is* connected to a network.

IV. Procedure

Reminder: This is a *viewing* activity. Do not change, disable, or reset any network setting. Close every dialog with **Cancel**. Record values exactly as Windows shows them.

Part A – Windows 11 Settings

1. Press **Windows + I** to open Settings, then click **Network & internet**. Write down the connection status shown at the top of the page (for example, "Not connected").
2. Click **Advanced network settings**. Under **Network adapters**, note every adapter listed (for example, Ethernet, Wi-Fi, Bluetooth Network Connection) and whether each one is enabled or disconnected.
3. Scroll down and click **Hardware and connection properties**. For each adapter, record its **Name**, **Description**, **Physical address (MAC)**, and **Status** in **Table 1**.

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Part B – Control Panel (Network Connections)

4. Press **Windows + R**, type **ncpa.cpl**, and press **Enter**. This opens the classic **Network Connections** window. A red X on an icon means the adapter has no network connection.
5. Right-click your **Ethernet** adapter and choose **Properties**. List the items checked under "This connection uses the following items" in **Table 2**.
6. Select **Internet Protocol Version 4 (TCP/IPv4)** and click **Properties**. Record whether the computer is set to **Obtain an IP address automatically** (DHCP) or **Use the following IP address** (static), and the same for DNS. Click **Cancel**, then **Cancel** again. Do not change anything.

Part C – Command Prompt (CMD)

7. Press the **Windows key**, type **cmd**, and press **Enter**. Run each command below, one at a time, and record the results in **Table 3**.

   hostname

   getmac /v

   ipconfig /all

   Tip: For a disconnected adapter, **ipconfig /all** shows "Media disconnected" and no IPv4 address, but it still shows the Description and Physical Address.
8. Test the computer's own TCP/IP software using the loopback address. Record whether you got replies and the time. For **ping localhost**, note the address that replied.

   ping 127.0.0.1

   ping localhost
9. Now observe what happens when there is no network. Record the **exact** message each command displays.

   ping 8.8.8.8

   nslookup google.com
10. View the ports your computer is listening on and its routing table. In Table 3, record three LISTENING local addresses with their port numbers, and the Netmask shown for the Network Destination 127.0.0.0.

    netstat -an | findstr LISTENING

    route print -4
11. Run the command below. It may take a few seconds. Scroll to **Network Card(s)** and record how many network cards are installed and the status of each.

    systeminfo

Part D – PowerShell

12. Right-click the **Start** button and choose **Terminal** (or press the **Windows key**, type **PowerShell**, and press **Enter**). Run each command below and record the results in **Table 4**.

    Get-NetAdapter

    Get-NetIPAddress -AddressFamily IPv4 | Format-Table InterfaceAlias, IPAddress, PrefixLength

    Test-Connection 127.0.0.1 -Count 4

    Get-NetTCPConnection -State Listen | Select-Object -First 5 LocalAddress, LocalPort

    Tip: PowerShell commands follow a Verb-Noun pattern (Get-NetAdapter = "get the network adapters"). Compare each result with the matching CMD command from Part C.

Part E – Analyze a Networked Computer

13. The output below was taken from a laboratory computer that **is** connected to a network. Study it, then complete **Table 5**.

Sample Output 1. ipconfig /all (excerpt) from a networked PC

```
Host Name . . . :                 CICT-LAB-PC07
Ethernet adapter Ethernet:
Description . . . :               Realtek PCIe GbE Family Controller
Physical Address . . . :          3C-52-82-A1-7F-09
DHCP Enabled . . . :              Yes
Link-local IPv6 Address . . . :   fe80::8c2d:4a1f:93b2:61e5%12(Preferred)
IPv4 Address . . . :              192.168.10.37(Preferred)
Subnet Mask . . . :               255.255.255.0
Lease Obtained . . . :            Wednesday, October 14, 2026 8:02:15 AM
Lease Expires . . . :             Thursday, October 15, 2026 8:02:15 AM
Default Gateway . . . :           192.168.10.1
DHCP Server . . . :               192.168.10.1
DNS Servers . . . :               192.168.10.1, 8.8.8.8
```

Part F – Design a Network for Our Laboratory

14. Our laboratory does not yet have a network. In the box after Table 5, draw a simple **star topology** network for it. Include at least **four PCs**, one **switch**, one **router** (default gateway), one **printer**, and the **ISP / Internet**. Use lines to show connections.

15. Assign each device a private IPv4 address from the network **192.168.10.0/24**, using **192.168.10.1** for the router. Label each device in your drawing and complete the **IP Address Plan** (Table 6).

Part G – Optional / Bonus

16. In CMD, run **type C:\Windows\System32\drivers\etc\hosts** to *view* (not edit) the hosts file. In one or two sentences, explain what this file is for and how it relates to DNS.

V. Data and Findings

Table 1. Network Adapters in Settings (Part A)

Connection status shown on the Network & internet page: _____

Adapter Name	Description	Physical Address (MAC)	Status
Ethernet	Realtek PCIe GbE Family Controller	d8:43ae:78:63:96	Not operational
Ethernet 2	VirtualBox Host-Only Ethernet Adapter	0a:00:27:00:00:0c	Operational
Ethernet 0	Intel(R) 82574L Gigabit Network	00:0c:29:7a:af:36	Not Present
Ethernet 3	Remote NDIS based Internet Sharing Device	4a:d7:1e:6f:bf:c4	Not Present

Table 2. Adapter Properties (Part B)

Items Checked Under "This connection uses the following items"	Item Checked (continued)
Client for Microsoft Networks	Allows your computer to access resources on a Microsoft network.
File and Printer Sharing for Microsoft Networks	Allows other computers to access resources on your computer using a Microsoft network

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Items Checked Under "This connection uses the following items"	**Item Checked (continued)**
Internet Protocol Version 6 (TCP /IPv6)	TCP/IP version 6. The latest version of the internet protocol that provides communication across diverse interconnected networks.
QoS Packet Scheduler	Quality of Service Packet Scheduler. This component provides network traffic control, including rate-of-flow and priotization services.

TCP/IPv4 Setting	**Automatic or Static?**
IP address	
DNS server address	255.255.255.0

Table 3. Command Prompt Results (Part C)

Command / Item	**Your Findings**	**OSI Layer**
hostname	DESKTOP -5Q638T3	Layer 7: Application
getmac /v: MAC address and Transport Name	Physical Address- C8-CB-9E-B7-99-BD Transport Name- \Device\Tcpip_{E84EA637-9863-4649-9F2E-C1F755EDODDE}	Layer 2: Data Link
ipconfig /all: Media State of Ethernet	Media Disconnected	—
ipconfig /all: DHCP Enabled	Yes	—
ping 127.0.0.1: replies and time	Reply from 127.0.0.1: bytes=32 times<1ms TTL=128	Layer 3: Network
ping localhost: address that replied	Reply from ::1: time<1ms	—
ping 8.8.8.8: exact message	Reply from 8.8.8.8 with 32 bytes of data: Request timed out.	Layer 3: Network

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Command / Item	Your Findings	OSI Layer
nslookup google.com: exact message	Server : globebroadband.net Address: fe80::8045:c8ff:feb8:3369 Non-authoritive answer: Name: google.com Addresses: 2a00:1450:4009:c0b::71 2a00:1450:4009:c0b::8a 2a00:1450:4009:c0b::86 142.251.29.102 142.251.29.138 142.251:29:101 142.251:29:100 142.251:29:139 142.251.29.113	Layer 7: Application
netstat -an: three LISTENING addresses:ports	Local Address: 0.0.0.0:135 Foreign address: 0.0.0.0:0 State: LISTENING Local Address: 0.0.0.0:445 Foreign Address: 0.0.0.0:0 State: LISTENING Local Address: 0.0.0.0:5040 Foreign Address: 0.0.0.0:0 State: LISTENING	Layer 4: Transport
route print -4: Netmask for 127.0.0.0	255.0.0.0	–
systeminfo: number of NICs and status	1 NIC(s) Installed. Status: Wifi Connected	–

Table 4. PowerShell Results (Part D)

Command	What It Showed	Matches CMD? (Y/N)
Get-NetAdapter: Name, Status, MacAddress, LinkSpeed	Name- Wi-Fi Status- Up MacAddress- C8-CB-9E-B7-99-BD LinkSpeed- 866.7 Mbps	Y
Get-NetIPAddress: IPv4 addresses and PrefixLength	IPv4: Address family PrefixLength: 16	Y
Test-Connection 127.0.0.1: result	DESKTOP -5Q... 127.0.0.1 192.168.254.196 fe80::f0f9:6bd3:62fe:5a05%17 32 0	Y
Get-NetTCPConnection -State Listen: two LocalAddress:LocalPort pairs	LocalAddress: :: :: LocalPort pairs: 63780 49673	Y

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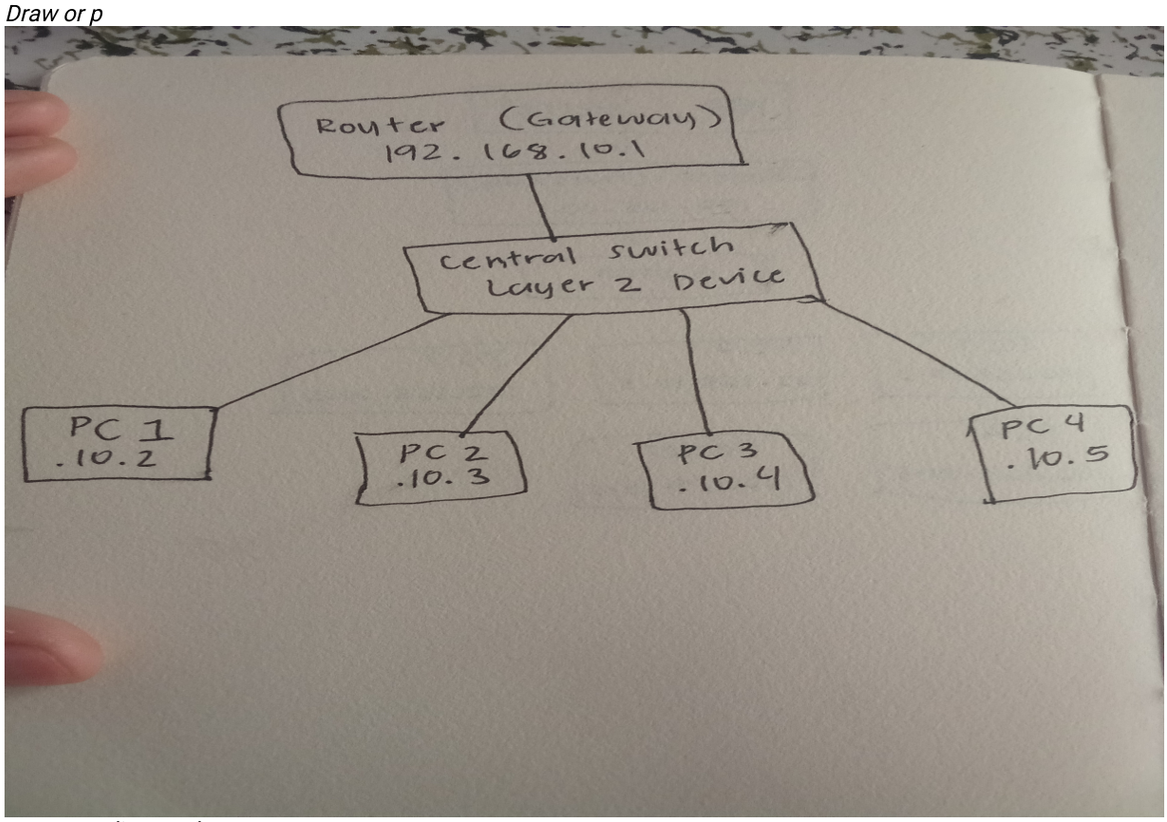
Table 5. Sample Output Analysis (Part E)

Question	Your Answer
IPv4 address of PC07	192.168.254.196(Preferred)
Subnet mask (and its /prefix)	255.255.255.0
Default gateway, and what device it usually is	192.168.254.254
DNS servers	fe80::8045:c8ff:feb8:3369%17 192.168.254.254
Was the address assigned by DHCP or set manually? How do you know?	The address was assigned by DHCP if ipconfig /all shows DHCP Enabled: Yes. If it shows No, the address is usually configured manually.
Public or private? Which RFC 1918 range?	Private IP address. It belongs to the RFC 1918 range 192.168.0.0 – 192.168.255.255 (also written as 192.168.0.0/16).
IPv4 address of PC07 in binary (four octets)	IP Address: 192.168.254.196 Binary Format: 11000000.10101000.11111110.11000100
MAC address, and the OSI layer where it is used	The MAC address is listed in ipconfig /all as Physical Address. It is used at Layer 2 – Data Link of the OSI model.
How long is the DHCP lease?	Thursday, 15 October 2026 7:53:33 pm Saturday, 17 October 2026 8:38:43 am
If the DHCP server failed, what kind of address would PC07 assign itself?	PC07 may assign itself an APIPA (Automatic Private IP Addressing) address in the range 169.254.0.0 – 169.254.255.255 (169.254.x.x) if it cannot obtain an IPv4 address from the DHCP server.

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Figure 1. Proposed Laboratory Network Diagram (Part F)

Draw or p



aste your diagram here.

Table 6. IP Address Plan (Part F)

Device	IPv4 Address	Subnet Mask	Default Gateway
Router (gateway)	192.168.10.1	255.255.255.0	—
PC 1	192.168.10.2	255.255.255.0	192.168.10.1
PC 2	192.168.10.3	255.255.255.0	192.168.10.1
PC 3	192.168.10.4	255.255.255.0	192.168.10.1
PC 4	192.168.10.5	255.255.255.0	192.168.10.1
Printer	192.168.10.6	255.255.255.0	192.168.10.1

Part G Notes (Optional / Bonus)

VI. Realization (Reflection)

In two to four sentences each, answer the following in your own words.

1. Even without a network connection, what network information did your computer still have? Why do

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you think it still has it?

Even without an active internet connection, a computer retains its permanent hardware configuration data, such as its MAC address (physical address) and local loopback address (127.0.0.1). It will also keep any statically assigned IP configurations, subnet masks, and cached DHCP settings stored in the operating system's registry. The computer keeps this data because these properties are tied to the local network interface card (NIC) and system software, meaning they do not require an active external connection to exist.

2. Compare Settings, Control Panel, CMD, and PowerShell. Which tool did you find most useful for viewing network information, and why?

Settings and Control Panel provide user-friendly graphical interfaces (GUI) that are great for quick, visual status checks, though Control Panel offers deeper legacy configurations. On the other hand, CMD and PowerShell are powerful command-line interfaces (CLI) that display comprehensive, raw data instantly without forcing you to click through multiple menus. CMD (using the ipconfig /all command) is often the most useful for viewing information because it quickly lists every adapter's IP, MAC address, and DNS status on a single, easy-to-read screen.

3. What did the messages from ping 8.8.8.8 and nslookup google.com tell you? How would you use messages like these when troubleshooting a real network?

The message from ping 8.8.8.8 verifies basic IP layer connectivity to a remote server, while nslookup google.com confirms that your DNS (Domain Name System) resolution is working correctly. When troubleshooting a live network, you use these tools to isolate the exact point of failure. For example, if you can successfully ping an IP address but cannot load a website or resolve its name via nslookup, you instantly know the problem lies with the DNS server rather than the physical internet connection.

4. Why should you be careful about sharing your MAC address, IP address, or ipconfig /all screenshots online? Relate your answer to data privacy.

Sharing screenshots of ipconfig /all, IP addresses, or MAC addresses online exposes unique hardware identifiers and network configurations to the public. Malicious actors can exploit this data to track your physical device, execute MAC spoofing attacks, or target your router with localized cyber threats like DDoS attacks. Protecting this information is crucial for data privacy because it acts as a digital footprint that map out your internal network defense layout.

VII. Scoring Guide

Criteria	Description	Points
Completeness	All tables (1–6) and required fields are filled out.	25
Accuracy	Recorded values match what the tools show and are consistent across Settings, CMD, and PowerShell; sample analysis, binary conversion, and OSI layers are correct.	30
Network Design	Diagram shows a star topology with PCs, switch, router, printer, and ISP; IP plan uses valid, non-duplicate addresses in 192.168.10.0/24.	15
Realization / Reflection	Answers are thoughtful, based on the student's own results, and well written.	20

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Criteria	Description	Points
Neatness & Timeliness	Worksheet is legible/well-formatted and submitted on time.	10
Total		**100**